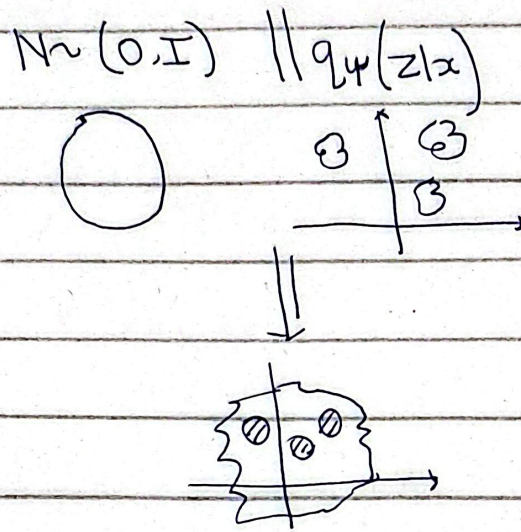


Lecture 5

→ how z learn mapping



Formal introduction to domain & domain shift

domain : context in which we define learning in terms of distribution

$$\mathcal{D} : \{ p(x), p(y|x), \underbrace{L}_{\text{set of loss function}} \}$$

→ Task : - within domain

- define our prediction function

$$\mathcal{T} : \{ p(x), p(y|x), \underbrace{l_j}_{\text{One loss function}} \}$$

SDST : Single domain Single Task

Input = MNIST $\rightarrow P(x), P(y|x)$

Task : Digit Recognition (10-class classification)

- Simple train through cross-entropy

MDST : Multiple domain Single Task

Input : PACs

$\hookrightarrow$ 4 domains

$\hookrightarrow$ Images from real world

$\hookrightarrow$ From cartoon

$\hookrightarrow$ From art

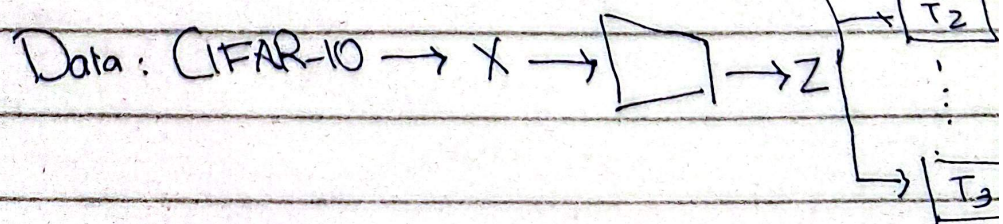
$\hookrightarrow$ From painting

Task : Image classification

$\downarrow$

each domain has 1 classes same

SDMT : Single Domain Multiple Task

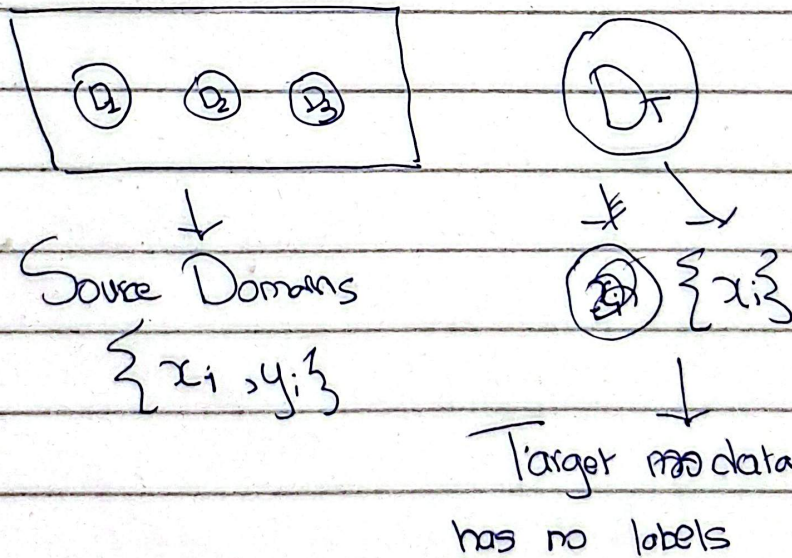


$T_1 = \text{Image Segmentation} \rightarrow L_1$
 $T_2 = \text{Object Detection} \rightarrow L_2$
 $T_n = \text{task Learning} \rightarrow L_n$

Each task function is trained on their own separate loss functions

* Transfer learning:

1) UDA: Unsupervised domain adaptation

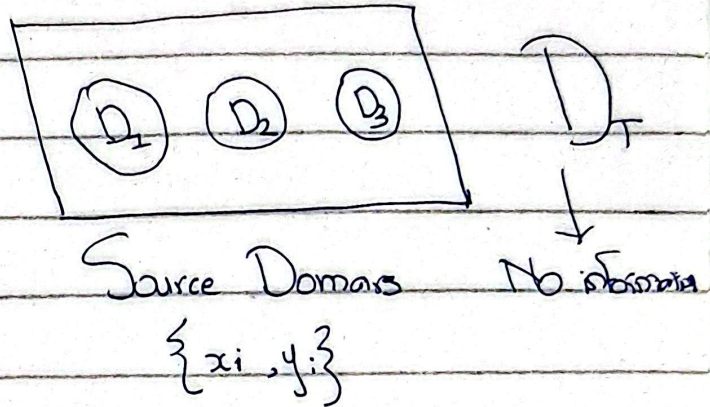


Train $\swarrow$ $\searrow$ Test
 Task: Learn target data's labels using both
 source domains and ~~domain~~ target domain
 Predict labels

UDA: Using target data, we might learn some semantics

DG: No information - Train only using source domains

2// Domain Generalization:



* Assumption:

Task is same i.e. Classification

1// ^{Setup} UDA: [Unsupervised domain adaptation]

D_S

1 source domain

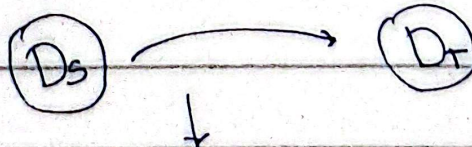
$P_S(x), P_S(y|x)$

D_T

Target Domain

$D_T(x), D_T(y|x)$

Domain Shift:



1. Covariate shift: $P_T(y|x) = P_S(y|x)$
 $P_T(x) \neq P_S(x)$

* Suppose $D_S = \{\text{cat, dog, lizard}\}$

↓
Assuming they have some labels

↓
 $D_T = \{\text{cat, dog, lizard}\}$ domain input distribution

↳ only domain can be different

Semantic Shift:

like D_S - real images

D_T = paintings

$$2_{\parallel} P_T(Y|X) \neq P_S(Y|X) \\ \Rightarrow P_S(X) \neq P_T(X)$$

in case



1. Closed Set DA (CSDA)

$Y_S = Y_T$ Covariate Shift

Diff level

1

2. Partial Set DA (PSDA)

$Y_T \subset Y_S$ Covariate + Semantic Shift

2

3. Open Set DA (OSDA)

$Y_S \subset Y_T$ $C_S \neq C_T$

3

↓
Target has classes
model has never
seen

Open Set Recognition $\rightarrow$ OSDA

4. Universal DA (UDA) $Y_S \cap Y_T \neq \emptyset$ | 4

$\Rightarrow$ Why our traditional DL fail

$\Downarrow$

ERM Rule:
$$\hat{P}_{\text{emp}}(f_0) = \frac{1}{N} \sum_{i=1}^N L_{\text{CE}}(f_0(x_i), y_i)$$

$\Downarrow$
Empirical risk minimization

$\Downarrow$
True Risk

$$P_{\text{True}}(f_0) = E_{P_S(x,y)} \left[L_{\text{CE}}(f_0(x), y) \right]$$

$\downarrow$
True is on true distribution of x, y

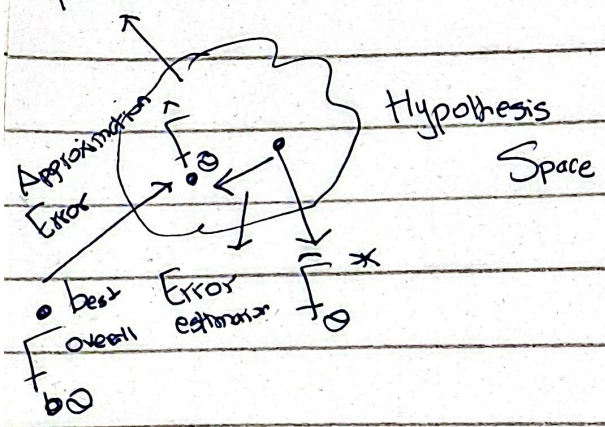
if empirical $\downarrow$

because its not available.

we minimize sample's empirical ~~function~~

by taking each sample's loss and averaging it

ResNet-152



- maybe my hypothesis space is wrong

- F_0 - chosen model in hypothesis space
- F_0^* - best model in hypothesis space
- $F_0^{\text{best overall}}$ - maybe my hypothesis space is wrong and there exists a best model outside of my hypothesis space

- estimation error [EE] = loss between $\hat{F}_0$ & $\hat{F}_0^*$
 $\hookrightarrow$ to be reducible or can be minimized

- ~~api~~
 - approximation error - loss between $\hat{F}_0$ & $\hat{F}_0^{\text{best}}$
 $\downarrow$
 irreducible

$$-EE - EE \rightarrow 0$$

$$N \uparrow, (x_i, y_i) \sim P_S(x_i, y_i)$$

where ① $N \rightarrow \infty$

② model capacity

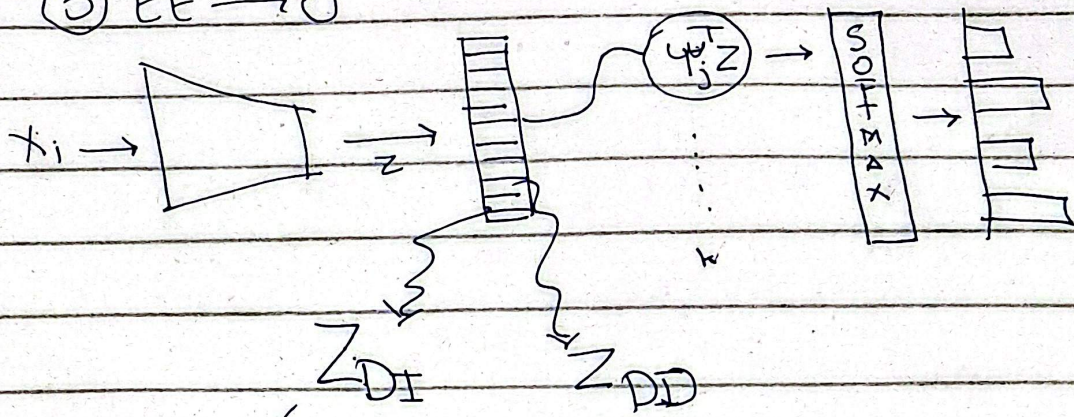
$\hookrightarrow$ Overblown

(not overfitting)

- model capacity can be measured through norm of model weights

Overfitting $EE \rightarrow 0$

③ $EE \rightarrow 0$



Z_{DI}

Z_{DD}

domain invariant

domain dependent

cat's dominant features

if something is common,

doesn't matter where it is sitting

like in all pictures, cat is

sitting on a table, model learns

table features too

Spurious Correlation:

correlation between Z_{DD} & Y_{label}

Ben-david bound:

$$L_T(f_0) \leq L_S(f_0) + \text{div} \left(\begin{matrix} P_S, P_T \\ \mathcal{Z} \end{matrix} \right)$$

$+ \min(a, b)$

$$a = E_{P_S} [P_S - P_T] \quad b = E_{P_T} [P_S - P_T]$$

R.H.S

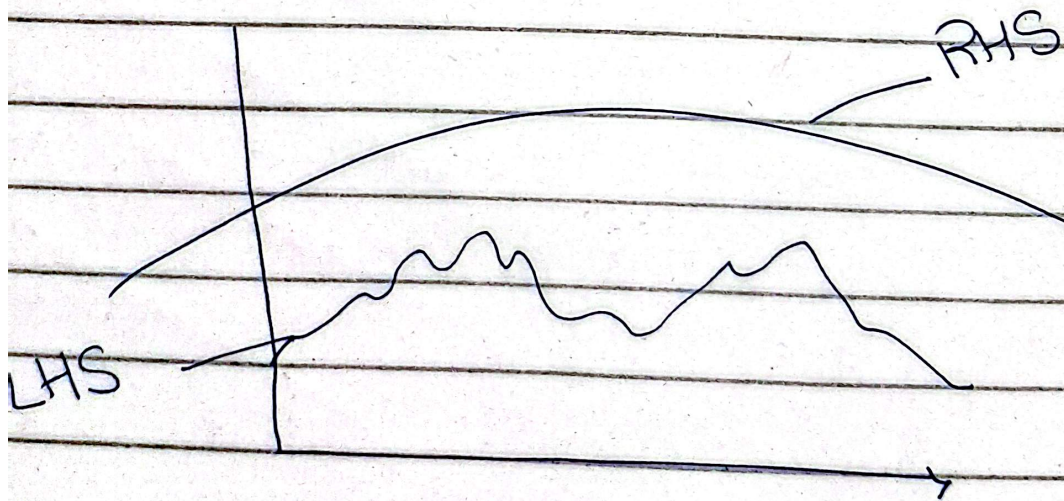
Ben-David bound:

$$L_T(f_0) \leq L_S(f_0) + d_{TV}(P_S(x), P_T(x))$$

$$+ \min(a, b)$$

$$a = E_{P_S} \left[|P_S(y|x) - P_T(y|x)| \right]$$

$$b = E_{P_T} \left[|P_S(y|x) - P_T(y|x)| \right]$$



→ if D_S and D_T are closer, that means
div distance between them is less

For closed set:

$$P_S(y|x) = P_T(y|x)$$

$$\min(a, b) = 0$$

* if $\text{div}(P_S(x), P_T(x)) \uparrow$, then

P_S Domains

and Domain T have

lot of distance b/w them

* $\min(a, b)$ — basically for difference in bbls

↳ open shift, universal shift

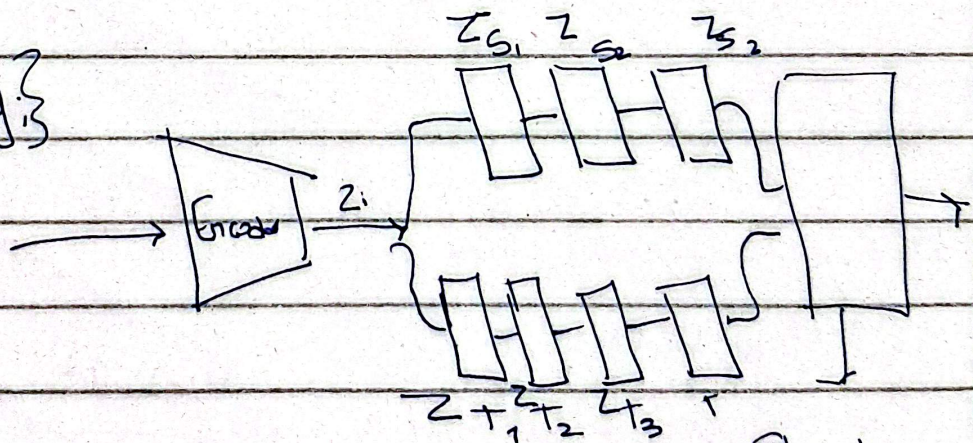
↓ RHS ↓, $L_S(f_0)$ ↓ } alignment

or make

$$\text{div}(P_S(x), P_T(x)) \downarrow$$

$D_S\{x_i, y_j\}$

$D_T\{x_i\}$

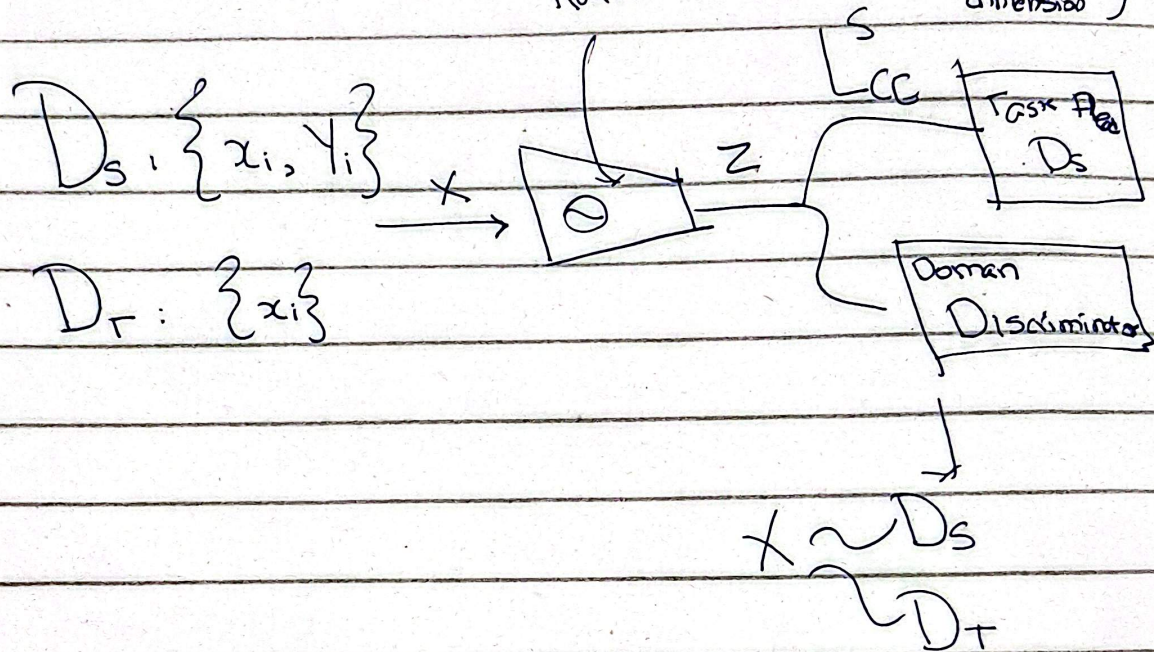


* we try to match

$$Z_S^1 = Z_T^1$$

$$L_T = L_{CE}^S + \lambda \sum_{i=1}^L \text{div}(Z_i^S, Z_i^T)$$

2, Adversarial Learning:



- we have to train encoder (e) so that z_i can fool Domain discriminator can distinguish whether x belong to D_S or D_T

↳ Thus, z_i belong to both D_S and D_T

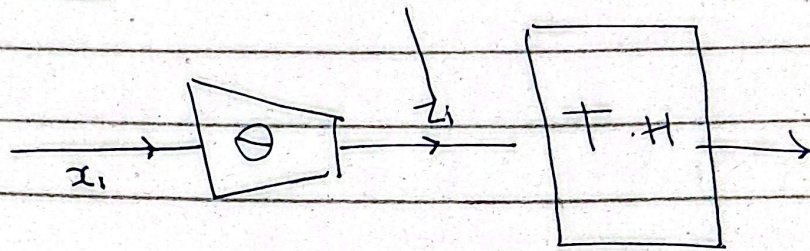
↓
This further means domain dependent features are ~~eliminated~~ there (confirm this)

3.11

* Contrastive :

$D_s \{x_i, y_i\}$

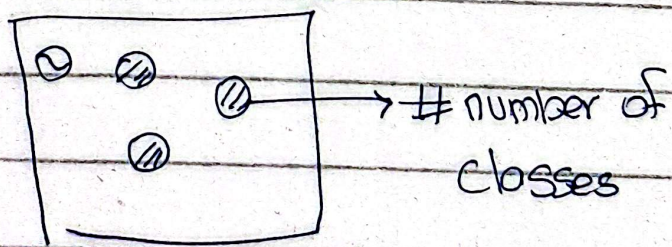
$D_r \{x_i\}$



$$\frac{e^{\sin(z, z^+)}}{e^{\sin(z, z^-)}}$$

1. basically we ignore $D_r \{x_i\}$, and train the whole ~~cluster~~ Contrastive model on $D_s \{x_i, y_i\}$,

2. When we are passing it through D_r encoder and z_i 's visualization shows



Pass ↓